Background

In this practical, you will fit a Bayesian hierarchical diagnostic test accuracy (DTA) model using R and Stan, through the `rstan` package.

The dataset

A systematic review assessed the diagnostic accuracy of computed tomography (CT), magnetic resonance imaging (MRI), and bone scintigraphy (BS) for clinically suspected scaphoid fractures in patients with normal initial radiographs. (Mallee et al., 2015, PMID 26045406). The review focused on the direct and indirect comparisons of these three imaging tests, with summary sensitivity and specificity as the main outcomes. Four reference standards were considered: scaphoid plain radiograph series conducted six to 14 weeks after the initial injury, use of two index tests, clinical findings often combined with an index test or repeated radiographs obtained after six weeks, and use of only one of the second-line modalities.

The dataset `26045406.xlsx` includes the following variables:

Variable name	Description
`study_id`	Study ID
`Study name`	Full Study Name
`year`	Year of Publication
`test_id`	Test ID (1: MRI, 2: BS, 3: CT)
`tp, fp, fn, tn`	True Positive, False Positive, False Negative, True Negative

Data requirements

The data contain diagnostic accuracy information from several studies and tests. The dataset should contain one row for each study-test 'observation' and include:

- A study identifier (e.g., in the example dataset, `study_id`)
- A diagnostic index test identifier (e.g., in the example dataset, `test_id`)
- The number of true positives (tp)
- The number of false negatives (fn)
- The number of true negatives (tn)
- The number of false positives (fp)

From these values, we then can derive the required:

- Diseased participants: $tp + fn$.

- Non-diseased participants: $tn + fp$.

Analysis steps

1. **Install and load the required R packages.**
   We will need `rstan`, as well as additional packages for importing the dataset and saving the results, such as `readxl` for Excel files and `writexl` for exporting results to Excel.

2. **Import the dataset.**

Make sure that the dataset includes, at a minimum, the information described above: a study identifier, an index-test identifier, and the corresponding 2×2 contingency table for each test evaluated in each study: true positives (tp), false negatives (fn), true negatives (tn), and false positives (fp).

3. **Prepare the data in the format required.**

Prepare the data as a named list, with each element corresponding to a variable required by the Stan model. The list should include the number of observations (`N`), the number of diagnostic tests (`Nt`), the number of studies (`Ns`), the numbers of true positives (`TP`) and true negatives (`TN`), the total numbers of diseased (`Dis=tp+fn`) and non-diseased (`Ndis=tn+fp`) participants, and the corresponding study (`study_id`) and diagnostic-test (`test_id`) identifiers.

> ‼ *Note: Before running the model, make sure that `study_id` and `test_id` are numeric identifiers beginning at 1, as they are used as Stan indices.*

4. **Fit the Bayesian DTA-NMA model (ANOVA model by Nyaga et al.)**

Set the working directory to the folder containing the Stan model (stan extension file), then fit the model using the prepared data list. You may begin with a short run to check that the model works before using the final settings.

The `warmup` iterations are used for sampler adaptation, while `iter` specifies the total number of iterations per chain. The `chains` argument determines the number of independent Markov chains, and `thin` retains every specified draw. After confirming that the model runs correctly, increase the number of iterations for the final analysis. All these settings are specified within the `stan()` function, which should include the path to the Stan model file and the data list created in Step 3.

5. **Examine the posterior estimates and convergence diagnostics.**
Examine the posterior estimates by reporting the posterior median (50% quantile) and the corresponding 95% credible interval (2.5%-97.5% quantiles). Assess convergence using the R-hat statistic and trace plots. Values of R-hat close to 1 and well-mixed chains indicate satisfactory convergence.

6. **Export the results and save the fitted model.**
You can export (a subset of or all) the results to Excel, CSV, or another suitable format using packages such as `writexl`, `readr`, or `openxlsx`, or save the complete fitted model as an `.RData` file.

7. **Results presentation.**
League tables for sensitivity and specificity separately, summaries of the network, diagnostic odds ratio and forest plots can be used to explore the results. Use the available functions (`mu_table`, etc) to generate these tables and graphics.

Questions

1. **How many patients were assessed in total?**

2. **What's the sensitivity and specificity for CT?**

3. **What's the relative specificity of CT vs MRI?**

4. **According to DOR, which test was ranked as best?**

5. Complete the league table

	(1) BS	(2) MRI	(3) CT
(1) BS			
(2) MRI			
(3) CT			

References

1. Mallee WH, Wang J, Poolman RW, Kloen P, Maas M, de Vet HC, Doornberg JN. Computed tomography versus magnetic resonance imaging versus bone scintigraphy for clinically suspected scaphoid fractures in patients with negative plain radiographs. Cochrane Database Syst Rev. 2015 Jun 5;2015(6):CD010023. doi: 10.1002/14651858.CD010023.pub2. PMID: 26045406; PMCID: PMC6464799.
2. Nyaga VN, Aerts M, Arbyn M. ANOVA model for network meta-analysis of diagnostic test accuracy data. Stat Methods Med Res. 2018 Jun;27(6):1766-1784. doi: 10.1177/0962280216669182. Epub 2016 Sep 20. PMID: 27655805.